

# Cluster 18121

To view the latest Map of Science clusters and data, visit the live Map at [sciencemap.eto.tech](https://sciencemap.eto.tech).

## Subjects

**Research fields:** Computer science, Mathematics, Engineering

The most common research fields of articles in this cluster. [Read more.](#)

**Research subfields:** Computer hardware, Algorithm, Electronic engineering

Common research subfields of articles in this cluster. [Read more.](#)

**Key concepts:** Neural networks, Activation Function, MLP circuit, Artificial Neural, feed-forward neural network

These key concepts are identified algorithmically using article titles and abstracts. [Read more.](#)

**37.50% of all articles are AI-related**

Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

**0.00% of English-language articles are related to robotics**

Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

**4.17% of English-language articles are related to computer vision**

Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

**0.00% of English-language articles are related to natural language processing**

Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

**0.00% of English-language articles are related to AI safety**

Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. [Read more.](#)

**0.00% of English-language articles are related to cybersecurity**

## Vital signs

**26** recent articles in the cluster

26 articles in this cluster were published in the five year period ending in 2023.

Average article is **28.66** years old

The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **73.00th** percentile

Over the three year period ending in 2023, this cluster grew faster than roughly 73.00% of other clusters.

Citation rating: **31.66th** percentile

A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year. In this case, the average article in this cluster is cited more often than roughly 31.66% of all other articles published in the same year.

**88.46%** of articles are in English

**Not expected** to grow extremely fast

As determined by ETO's growth prediction model. [Read more.](#)

# Countries and languages

## Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 21 of 26 articles. 20 of these articles had at least one author from an organization in these ten countries.

| Country       | Articles (five year period ending in 2023) | Yearly citations per article (average) |
|---------------|--------------------------------------------|----------------------------------------|
| United States | 4                                          | 0.47                                   |
| France        | 4                                          | 0.48                                   |
| Japan         | 3                                          | 1.20                                   |
| India         | 3                                          | 0.69                                   |
| China         | 3                                          | 0.07                                   |
| Indonesia     | 2                                          | 0.13                                   |
| Canada        | 1                                          | 1.75                                   |
| Italy         | 1                                          | 1.67                                   |

## Top funding countries

Data available for 4 of 26 articles. 4 of these articles disclosed funding from one or more organizations in these ten countries.

| Country | Articles (last five years) | Yearly citations (average) |
|---------|----------------------------|----------------------------|
| Japan   | 2                          | 0.40                       |

# Articles and sources

## Core articles

Core articles are especially highly connected to other papers within the cluster.

[Implementation of Analog Perceptron as an Essential Element of Configurable Neural Networks, 2020](#)

2020: Switzerland). 4 citations.

[Implementation of RRAM based Swish Activation Function and its Derivative on 28nm FD-SOI](#)

2022: International Electrical Engineering Congress. 1 citations.

[Feed-Forward Neural Networks](#)

2021: Deep Learning with Python. 7 citations.

[Best Practices for Designing and Training Neural Networks Subjected to Random Variations](#)

2021: International Symposium on Circuits and Systems. 2 citations.

[Voltage-to-Voltage Sigmoid Neuron Activation Function Design for Artificial Neural Networks](#)

2022: Latin American Symposium on Circuits and Systems. 5 citations.

[Stone Carving Art Image Recognition Approach Based on Artificial Neural Network](#)

2020: 2020 5th International Conference on Smart Grid and Electrical Automation (ICSGEA). 1 citations.

[Analyzing the Effects of Noise and Variation on the Accuracy of Analog Neural Networks](#)

2020: Midwest Symposium on Circuits and Systems Conference Proceedings. 7 citations.

[An Analog CMOS Implementation for Multi-layer Perceptron With ReLU Activation](#)

2020: International Conference on Modern Circuits and Systems Technologies. 14 citations.

## Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

No data available.

## Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

[Implementation of Analog Perceptron as an Essential Element of Configurable Neural Networks, 2020](#)

2020: Switzerland). 4 citations.

[Feed-Forward Neural Networks](#)

2021: Deep Learning with Python. 7 citations.

[Best Practices for Designing and Training Neural Networks Subjected to Random Variations](#)

2021: International Symposium on Circuits and Systems. 2 citations.

[Voltage-to-Voltage Sigmoid Neuron Activation Function Design for Artificial Neural Networks](#)

2022: Latin American Symposium on Circuits and Systems. 5 citations.

[Stone Carving Art Image Recognition Approach Based on Artificial Neural Network](#)

2020: 2020 5th International Conference on Smart Grid and Electrical Automation (ICSGEA). 1 citations.

[NEURAL NETWORK ARCHITECTURES](#)

2023: Infokommunikacionnye tehnologii. 23 citations.

[Analyzing the Effects of Noise and Variation on the Accuracy of Analog Neural Networks](#)

2020: Midwest Symposium on Circuits and Systems Conference Proceedings. 7 citations.

[An Analog CMOS Implementation for Multi-layer Perceptron With ReLU Activation](#)

2020: International Conference on Modern Circuits and Systems Technologies. 14 citations.

## Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 25 out of 26 articles. 10 of these articles came from these sources.

| Source                                                                                                     | Articles (last five years) |
|------------------------------------------------------------------------------------------------------------|----------------------------|
| International journal of circuit theory and applications                                                   | 1                          |
| IEEE International Joint Conference on Neural Networks (IJCNN)                                             | 1                          |
| International Journal of Electrical and Electronic Engineering & Telecommunications                        | 1                          |
| 2020 5th International Conference on Smart Grid and Electrical Automation (ICSGEA)                         | 1                          |
| Engenharia: criação, construção e manutenção                                                               | 1                          |
| 2024 International Conference on Artificial Intelligence, Computer, Data Sciences and Applications (ACDSA) | 1                          |
| International Conference on Modern Circuits and Systems Technologies                                       | 1                          |
| Journal of Physics: Conference Series                                                                      | 1                          |
| Midwest Symposium on Circuits and Systems Conference Proceedings                                           | 1                          |
| Deep Learning with Python                                                                                  | 1                          |

# Authors, organizations, and funders

## Top authors

The authors with the most articles from the last five years in this cluster. Data available for 25 of 26 articles. 8 of these articles were written by these authors.

| Author                  | Articles (last five years) | Yearly citations (average) |
|-------------------------|----------------------------|----------------------------|
| Shigetoshi Nakatake     | 2                          | 3.60                       |
| David V. Anderson       | 2                          | 0.95                       |
| Devon Janke             | 2                          | 1.65                       |
| Chao Geng               | 2                          | 3.60                       |
| Qingji Sun              | 2                          | 3.60                       |
| Lars Hedrich            | 1                          | 0.00                       |
| Mohamad Aldy Rozikin    | 1                          | 0.25                       |
| Nikhil Ketkar           | 1                          | 1.75                       |
| Prasoon Ambalathankandy | 1                          | 0.00                       |
| Masayuki Ikebe          | 1                          | 0.00                       |

## Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 21 of 26 articles. 11 had at least one author from one of these ten organizations.

| Organization                                              | Articles (last five years) | Yearly citations (average) |
|-----------------------------------------------------------|----------------------------|----------------------------|
| The University of Kitakyushu - Japan                      | 2                          | 1.8                        |
| Georgia Institute of Technology - United States           | 2                          | 0.95                       |
| T  l  com Paris - France                                  | 1                          | 0.25                       |
| Birla Institute of Technology and Science, Pilani - India | 1                          | 0.33                       |
| Vancouver, Canada                                         | 1                          | 1.75                       |
| Universitas Jenderal Achmad Yani - Indonesia              | 1                          | 0                          |
| Lebanese International University - Lebanon               | 1                          | 0                          |
| Shandong Women’s University - China                       | 1                          | 0.2                        |
| University of Maryland, College Park - United States      | 1                          | 0                          |
| Centre National de la Recherche Scientifique - France     | 1                          | 0                          |

## Organization types

This table covers the 20 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 26 articles total). [Read more](#)

| Commercial | Education | Nonprofit | Government |
|------------|-----------|-----------|------------|
| 0.00%      | 51.92%    | 1.92%     | 1.92%      |

## Top industry organizations

No data available.

## Top funders

Data available for 5 out of 26 recent articles. 5 of these articles disclosed funding from one or more of these organizations.

| Funding organization                                                   | Articles (last five years) | Yearly citations (average) |
|------------------------------------------------------------------------|----------------------------|----------------------------|
| Ministry of Education, Culture, Sports, Science and Technology - Japan | 1                          | 0.80                       |
| VLSI Design and Education Center (VDEC) at the University of Tokyo     | 1                          | 0.00                       |
| Czech Technical University in Prague - Czech Republic                  | 1                          | 0.00                       |
| Cadence Design System, Inc.                                            | 1                          | 0.00                       |
| Ministry of Science and Technology - Vietnam                           | 1                          | 0.00                       |
| Strategic Information and Communications R&D Promotion Program (SCOPE) | 1                          | 0.00                       |
| internal CTU                                                           | 1                          | 0.00                       |
| NSF-ANR CHAMELEON project                                              | 1                          | 0.00                       |
| České Vysoké Učení Technické v Praze                                   | 1                          | 0.00                       |
| Japan Society for the Promotion of Science - Japan                     | 1                          | 0.80                       |

## Funder types

This table covers the 4 articles in this cluster from the last five years with data about their funding organization types (out of 26 articles total).

| Commercial | Education | Nonprofit | Government |
|------------|-----------|-----------|------------|
| 0.00%      | 40.00%    | 20.00%    | 40.00%     |

# Patents and industry

**346** patents cite articles in this cluster ( **94.8th** percentile among all clusters)

**8.33%** of articles in the cluster are cited by patents

**0%** of articles have at least one author from industry

## Top patent titles

| Patent title                                                                              | Same-cluster citations |
|-------------------------------------------------------------------------------------------|------------------------|
| SEMICONDUCTOR DEVICE AND ELECTRONIC DEVICE                                                | 7                      |
| SEMICONDUCTOR DEVICE AND SYSTEM USING THE SAME                                            | 4                      |
| INTEGRATE AND FIRE ELECTRONIC NEURONS                                                     | 4                      |
| SYSTEMS AND METHODS OF SPARSITY EXPLOITING                                                | 3                      |
| REGISTER AND ACTIVE ELEMENT MACHINES: COMMANDS, PROGRAMS, SIMULATORS AND TRANSLATORS      | 3                      |
| CONVOLUTIONAL NEURAL NETWORK                                                              | 3                      |
| NON-DETERMINISTIC SECURE ACTIVE ELEMENT MACHINE                                           | 3                      |
| POWER MANAGEMENT FOR AN ANALOG NEURAL MEMORY IN A DEEP LEARNING ARTIFICIAL NEURAL NETWORK | 3                      |
| AN ANALOGUE ELECTRONIC DEEP NEURAL NETWORK                                                | 3                      |

## Top patent assignees

| Patent assignee                                                              | Same-cluster citations |
|------------------------------------------------------------------------------|------------------------|
| MICROCHIP TECHNOLOGY (UNITED STATES)                                         | 9                      |
| インターナショナル・ビジネス・マシーンズ・コーポレーション<br>INTERNATIONAL BUSINESS MACHINES CORPORATION | 8                      |
| シリコン ストリージ テクノロジー インコーポレイテッド                                                 | 8                      |
| 실리콘 스토리지 테크놀로지 인크                                                            | 8                      |
| IBM (UNITED STATES)                                                          | 22                     |
| INTERNATIONAL BUSINESS MACHINES CORPORATION                                  | 21                     |
| SEMICONDUCTOR ENERGY LABORATORY CO. LTD                                      | 11                     |
| SHOPPING CLUB XIAOSONG PRODUCTION OFFICE                                     | 11                     |
| SELEZNIOVAS ALEKSANDRAS                                                      | 11                     |

# Connections

## Top citing clusters

| Cluster               | Number of significant citations | CSET extracted phrases                                                                                       |
|-----------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------|
| <a href="#">29730</a> | 1                               | Deep Neural Networks, Convolutional Neural Network, neural network models, Fault Injection, DNN accelerators |

## Top cited clusters

| Cluster               | Number of significant citations | CSET extracted phrases                                                                                                               |
|-----------------------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| <a href="#">20399</a> | 1                               | CMOS analog multiplier, proposed circuit, CMOS technology, power consumption, analog multiplier circuit                              |
| <a href="#">5519</a>  | 1                               | neural recording, CMOS process, input impedance, EEG recording system, neural signal                                                 |
| <a href="#">11174</a> | 1                               | Spiking Neural Networks, Deep Spiking Neural, artificial neural, deep SNNs, Neural Network based                                     |
| <a href="#">3255</a>  | 1                               | neuromorphic computing, artificial neural networks, synaptic devices, Optoelectronic Neuromorphic Device, devices based neuromorphic |
| <a href="#">29730</a> | 1                               | Deep Neural Networks, Convolutional Neural Network, neural network models, Fault Injection, DNN accelerators                         |
| <a href="#">11504</a> | 1                               | Breast Cancer, Machine Learning Algorithms, breast cancer classification, support vector machine, Wisconsin Diagnostic Breast        |
| <a href="#">25087</a> | 1                               | deep neural networks, softmax activation function, Neural Network Hardware, hardware implementation, Sigmoid Function                |
| <a href="#">5215</a>  | 1                               | SNDR, modulator, Delta Sigma modulators, ADC achieves, continuous-time delta-sigma ADC                                               |
| <a href="#">25159</a> | 1                               | SRAM CIM macros, energy efficiency, MAC operations, neural networks, RRAM based CIM                                                  |

## Top GitHub repositories

No data available.

The Map of Science runs on bibliometric data from [Clarivate Web of Science](#) and other sources. For details on our sourcing, methodology, and limitations, see the [tool documentation](#).